

Math 521, Section 2
Homework 11
Selected Solutions

Problem 2. Let (X, d) be a metric space.

- (a) Prove that if $F \subseteq X$ is disconnected and closed in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
- (b) Prove that if $U \subseteq X$ is disconnected and open in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Solution: (a) As F is disconnected, then by Lecture 29 we know that there exists a set $A \neq \emptyset, F$ that is both open and closed in F . Set $B = F \setminus A$. We will show that these sets A and B satisfy the desired properties.

- A and B are nonempty: since $A \neq \emptyset$ and $A \neq F$.
- A is closed in X : We know that A is closed in F . As F is closed in X , then A is also closed in X (by Lecture 24).
- B is closed in X : We know that A is open in F , and so $B = F \setminus A$ is closed in F . As F is closed in X , then B must also be closed in X .
- A and B are separated: We know that $A \cap B = \emptyset$, since $B = F \setminus A$. As A is closed in X , then $\overline{A} = A$ and so $\overline{A} \cap B = A \cap B = \emptyset$. Similarly, as B is closed in X , we have $A \cap \overline{B} = A \cap B = \emptyset$.

(b) As U is disconnected, then by Lecture 29 we know that there exists a set $A \neq \emptyset, U$ that is both open and closed in U . Set $B = U \setminus A$. We will show that these sets A and B satisfy the desired properties.

- A and B are nonempty: since $A \neq \emptyset$ and $A \neq U$.
- A is open in X : We know that A is open in U . As U is open in X , then A is also open in X (by Lecture 24).
- B is open in X : We know that A is closed in U , and so $B = U \setminus A$ is open in U . As U is open in X , then B must also be open in X .
- A and B are separated: We know that $A \cap B = \emptyset$, since $B = U \setminus A$. Then

$$\begin{aligned} A \cap B = \emptyset &\implies A \subseteq B^c \\ &\implies \overline{A} \subseteq \overline{B^c} = B^c \\ &\implies \overline{A} \cap B = \emptyset. \end{aligned}$$

(In the second line, we used the fact that B^c is closed in X , since B is open.) Similarly,

$$\begin{aligned} A \cap B = \emptyset &\implies B \subseteq A^c \\ &\implies \overline{B} \subseteq \overline{A^c} = A^c \\ &\implies A \cap \overline{B} = \emptyset. \end{aligned}$$

□

Problem 3. Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Solution: Suppose towards a contradiction that X is disconnected. Then there exist sets $A, B \subseteq X$ so that:

- $X = A \cup B$,
- A and B are nonempty,
- A and B are separated.

As A and B are nonempty, then there exists a point $a \in A$ and $b \in B$. The problem statement then tells us that there exists a connected set $C \subseteq X$ so that $a, b \in C$. As C is connected and C is contained in the union of the separated sets A and B , then by Lecture 29 we know that either $C \subseteq A$ or $C \subseteq B$. As $a \in A \cap C$, then we must be in the first case: $C \subseteq A$ and $B \cap C = \emptyset$. But this contradicts that $b \in B \cap C$. $\square$

Problem 4. Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Solution: Suppose towards a contradiction that $\overline{A}$ is disconnected. Then there exist sets $B, C \subseteq X$ so that:

- $\overline{A} = B \cup C$,
- B and C are nonempty,
- B and C are separated.

As A is connected, then by Lecture 29 either $A \subseteq B$ or $A \subseteq C$. We may assume that $A \subseteq B$ after swapping B and C if necessary. Then $\overline{A} \subseteq \overline{B}$. On the other hand, we know B and C are separated, and so $\overline{B} \cap C = \emptyset$. Together, we conclude that $\overline{A} \cap C = \emptyset$. But $\overline{A} = B \cup C$, and so the only way this can happen is if $C = \emptyset$. This contradicts that C is nonempty. $\square$

Problem 5. Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Solution: Suppose towards a contradiction that $A \cup B$ were disconnected. Then there exist sets $C, D \subseteq X$ so that:

- $A \cup B = C \cup D$,
- C and D are nonempty,
- C and D are separated.

As $A \subseteq C \cup D$ and A is connected, then either $A \subseteq C$ or $A \subseteq D$ by Lecture 29. We may assume that $A \subseteq C$ after swapping C and D if necessary.

Similarly, as $B \subseteq C \cup D$ and B is connected, then either $B \subseteq C$ or $B \subseteq D$. On the other hand, we know that $A \subseteq C$ and $A \cap B \neq \emptyset$, and so at least one point of B is in C . Together, we conclude that all of B is contained in C .

Altogether, we conclude $A \cup B = C$. This contradicts that D is nonempty. $\square$

Problem 6. Let (X, d) be a metric space.

- (a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
- (b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Solution: (a) As A and B are separated, then

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Similarly, as A and C are separated, we know

$$\overline{A} \cap C = \emptyset \quad \text{and} \quad A \cap \overline{C} = \emptyset.$$

Together, we see that

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C) = \emptyset \cup \emptyset = \emptyset,$$

and

$$A \cap \overline{(B \cup C)} = A \cap (\overline{B} \cup \overline{C}) = (A \cap \overline{B}) \cup (A \cap \overline{C}) = \emptyset \cup \emptyset = \emptyset.$$

(Recall that $\overline{B \cup C} = \overline{B} \cup \overline{C}$ by Lecture 23.) Therefore A and $B \cup C$ are separated.

(b) Suppose towards a contradiction that F is disconnected. Then there exist sets $A, B \subseteq X$ so that:

- $F = A \cup B$,
- A and B are nonempty,
- A and B are separated.

As $F \cap G$ is connected and $F \cap G$ is contained in the union of the separated sets A and B , then by Lecture 29 we know that either $F \cap G \subseteq A$ or $F \cap G \subseteq B$. We may assume that $F \cap G \subseteq A$ after swapping A and B if necessary.

Claim: B and G are separated. First, we note that $\overline{B} \cap G = \emptyset$, since

$$\begin{aligned} x \in \overline{B} &\implies x \notin A \quad \text{and} \quad x \in F \\ &\implies x \notin F \cap G \quad \text{and} \quad x \notin G \setminus F \\ &\implies x \notin G. \end{aligned}$$

Next, we use that G is closed to show $B \cap \overline{G} = \emptyset$:

$$\begin{aligned} x \in \overline{G} = G &\implies x \in G \setminus F \quad \text{or} \quad x \in F \cap G \subseteq A \\ &\implies x \notin B. \end{aligned}$$

Together, we conclude B and G are separated.

Therefore, applying part (a) to the sets B, A and B, G , we see that B and $A \cup G$ are separated. Note that B and $A \cup G$ are both nonempty since A and B are, and

$$B \cup A \cup G = F \cup G.$$

Altogether, we conclude that $F \cup G$ is disconnected, which is a contradiction. □